

Exercise 6 - Excited states and spectra of CFH₂-OH (fluoromethanol)

Comparison table

State	TDA E / eV	TDA λ / nm	TDA f	CIS E / eV	CIS λ / nm	CIS f
S1	8.10	153.1	0.0087	9.81	126.4	0.0082
S2	8.90	139.2	0.0193	11.05	112.2	0.0444
S3	9.43	131.5	0.0456	11.86	104.5	0.1865
S4	9.93	124.8	0.0144	12.43	99.7	0.0165
S5	10.10	122.8	0.0586	13.03	95.2	0.2335

UV/Vis spectrum

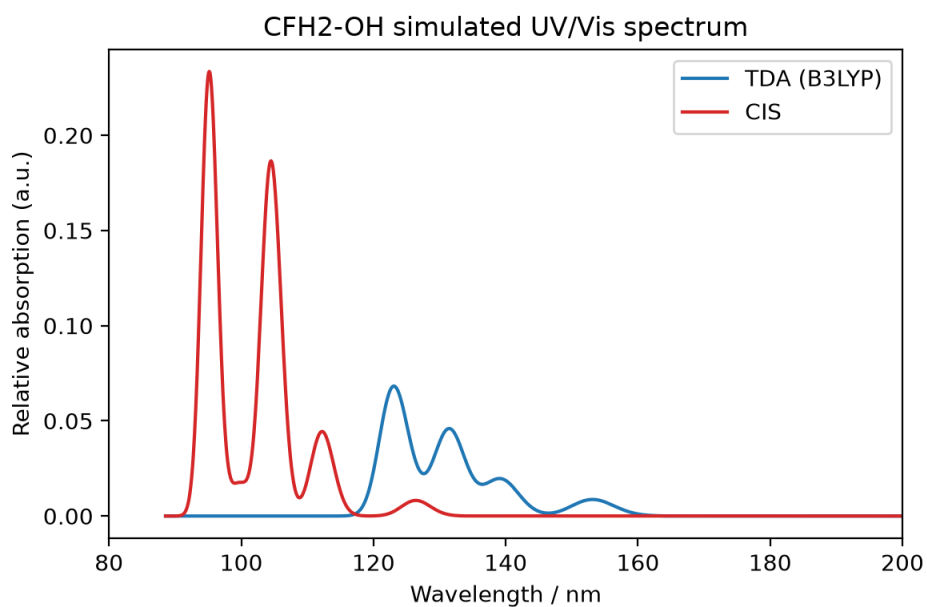


Figure 1. Simulated UV/Vis spectrum (Gaussian broadening, FWHM = 0.4 eV) from the TDA and CIS excitation energies and oscillator strengths above.

CIS gives systematically higher excitation energies than TDA/B3LYP across all five states (by roughly 1.7-3.0 eV), consistent with CIS lacking electron correlation.